

Study on Long-term Consistency and Persistent Behavior in Language Models via Recurrent Latent States

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Abstract

While Large Language Models (LLMs) demonstrate exceptional linguistic capabilities within localized contexts, their inference process remains fundamentally stateless. This architectural limitation prevents models from maintaining a continuously evolving internal representation across sessions, leading to the loss of user preferences, stylistic nuances, and long-term contextual dependencies. This research proposes the **Recurrent Latent State Module (RLSM)**, which introduces a low-dimensional state vector updated continuously over time. Regulated by a **prediction-error-driven gating mechanism**, the RLSM achieves a dynamic equilibrium between historical stability and environmental adaptability. Furthermore, we extend this mechanism into a lightweight closed-loop system, enabling models to transition from passive responders to entities with persistent behavioral patterns. This work aims to establish a feasible path between “stateless responders” and “continuous agents,” providing a novel architectural foundation for long-term interactive language models.

1 Introduction

The core computational process of modern LLMs is essentially conditional generation: the model produces output based on the current context but fails to sustain an internal state between independent sessions. While computationally efficient, this design imposes severe constraints in long-term scenarios. Specifically, user preferences struggle to accumulate across sessions, dialogue personas lack stability, and long-term dependencies must be repeatedly provided in the prompt.

Existing solutions like long-context modeling or RAG are essentially “input extensions” rather than “state formations.” This research addresses a fundamental question: *Can we introduce a continuously evolving internal state that allows LLMs to develop stable and adaptive behaviors across multi-session interactions?* We conceptualize this as a transition toward systems with continuous internal dynamics. By designing a pluggable recurrent module, we aim to grant models a minimal form of “internal continuity” without altering their primary transformer architecture.

2 Related Work and Problem Positioning

The current landscape of AI memory exhibits a structural gap. On one hand, stateless Transformers utilize context windows to process history, but this remains a re-encoding of inputs rather than a state update. On the other hand, LLM Agent systems utilize memory buffers or knowledge bases, which rely on discrete storage and retrieval, often lacking a unified, continuous state representation [1].

While Recurrent Neural Networks (RNNs) and State-Space Models (SSMs) like Mamba provide state mechanisms, they are primarily optimized for sequence modeling and typically reset their states at task boundaries, lacking the setting for cross-task continuous operation [2]. Furthermore, although prediction errors have long been used in cognitive modeling and reinforcement learning to drive exploration, there is a lack of systematic attempts to embed them directly into the state-update mechanisms of LLMs [3]. This research bridges these gaps by introducing a continuous, low-dimensional, and recursively updated internal state as the organizational foundation for long-term behavior.

3 Methodology

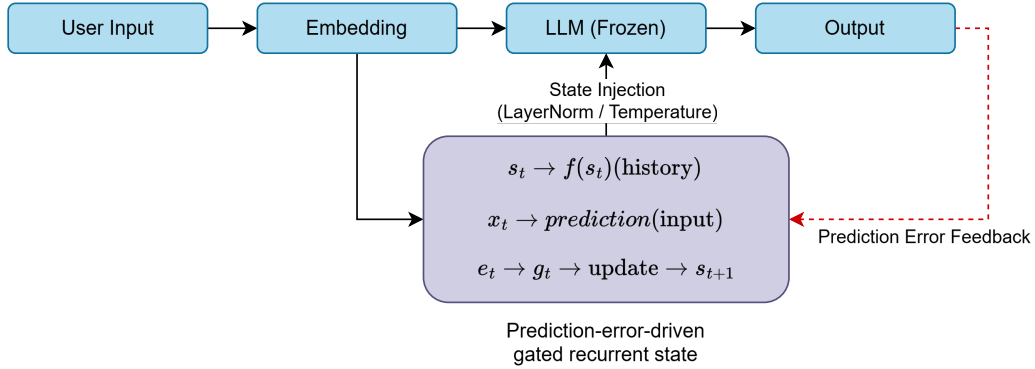


Figure 1: The architecture of the proposed Recurrent Latent State Module (RLSM). The system maintains a continuous latent state s_t , updated recursively based on the prediction error e_t . The state is then non-invasively injected into a frozen LLM via LayerNorm bias or attention temperature to ensure long-term behavioral consistency.

3.1 Persistent Latent State Representation and Update Logic

As illustrated in Figure 1, we introduce a latent state vector $s_t \in \mathbb{R}^N$ to encode cumulative information over time. Unlike context splicing, this state persists across sessions and serves as a latent variable for model behavior. The state is updated via the following recursive rule:

$$s_{t+1} = (1 - g_t) \cdot f(s_t) + g_t \cdot \mathbf{W}_{in} \mathbf{x}_t \quad (1)$$

where $\mathbf{x}_t$ represents the current input embedding, $f(\cdot)$ is a leaky non-linear evolution function—such as $f(s) = \tanh(\mathbf{W}s + \boldsymbol{\theta})$ —and $g_t \in [0, 1]$ is the gating coefficient. The state vector is never reset between sessions.

3.2 Prediction-Error-Driven Gating Mechanism

To prevent the state from being overly sensitive to input noise, the gating coefficient is determined by the standardized relative change in prediction error e_t . We define $e_t = \|\mathbf{x}_t - \hat{\mathbf{x}}_t\|^2$, where $\hat{\mathbf{x}}_t = \mathbf{M}_{pred} \cdot [\mathbf{s}_{t-1}; \mathbf{a}_{t-1}]$ and $\mathbf{a}_{t-1}$ is the previous action (output) embedding. The gate responds to the “surprisingness” of the error:

$$g_t = \beta g_{t-1} + (1 - \beta) \cdot \sigma \left(\alpha \cdot \frac{e_t - \mu_t}{\sqrt{\sigma_t^2 + \epsilon}} \right) \quad (2)$$

where μ_t and σ_t^2 are the exponential moving mean and variance of the error sequence. This ensures that when the environment is highly predictable, the state actively absorbs new information ($g_t \rightarrow 1$); if prediction fails, the system relies on historical evolution $f(s_t)$ to maintain stability ($g_t \rightarrow 0$).

3.3 Stability and Dynamical Analysis

Since the system operates over infinite time scales, we focus on boundedness, non-collapse, and input responsiveness. We analyze the system as a non-autonomous discrete dynamical system:

- **Global Attraction:** Utilizing the contraction mapping principle, by constraining the spectral radius $\rho(\mathbf{W}) \approx 0.95$, we ensure the state converges to a bounded invariant set.
- **Input-to-State Stability (ISS):** By treating external terms as bounded perturbations and assuming normalized input embeddings ($\|\mathbf{x}_t\| \leq 1$), a Lyapunov function $V(\mathbf{s}) = \|\mathbf{s}\|^2$ satisfies $\Delta V_t \leq -\alpha V_t + \gamma$, proving the state norm remains bounded.
- **Dimensional Non-collapse:** We add an orthogonality regularization term $\lambda \|\mathbf{W}^\top \mathbf{W} - \mathbf{I}\|_F^2$ to the update rule, preventing the state space from degrading into a low-dimensional manifold.

3.4 Localized Parameter Updates and Integration

To avoid the high cost of Backpropagation Through Time (BPTT), all parameters utilize **local online update rules**. The predictor $\mathbf{M}_{pred}$ is updated via the outer product of error and input (LMS rule). The recurrent weight $\mathbf{W}$ is driven by state correlations, while the action mapping $\mathbf{A}$ utilizes eligibility traces and prediction error changes as advantage signals to reinforce error-reducing action sequences.

Integration with the Transformer is achieved through two non-invasive schemes:

- **Scheme A (State-Conditioned LayerNorm Bias):** We map $\mathbf{s}_t$ to element-wise bias and scale terms for each Transformer layer’s normalization: $\text{LN}(\mathbf{x}) = \frac{\mathbf{x} - \mu}{\sigma} \odot (1 + \gamma_t^{(l)}) + \beta_t^{(l)}$ [4].
- **Scheme B (Latent-Guided Attention Temperature):** $\mathbf{s}_t$ is compressed into a scalar temperature coefficient τ_t acting on the attention Softmax, adjusting the model’s sharp or flat output distribution based on state uncertainty.

4 Experimental Framework

- **Stage I: Dynamical Property Validation.** Verify state norm stability, autocorrelation functions, and effective rank (via PCA spectrum) in a “Toy Grid World” simulation.
- **Stage II: Cross-Session Persona Consistency.** Conduct multiple independent dialogues where no context is shared. Consistency will be measured using Persona Consistency Scores and embedding similarity [5].
- **Stage III: User Preference Adaptation.** Measure how effectively the continuous state learns as a user progressively expresses preferences, tracking prediction accuracy and forgetting rates.
- **Stage IV: Closed-Loop Behavioral Analysis.** Evaluate behavioral diversity (entropy) and state-action mutual information to verify if the system develops persistent behavioral structures.

5 Timeline (12 Months)

- **M1–M3:** Module development and integration with frozen backbones.
- **M4–M6:** Execution of Stage I & II experiments; stability tuning.
- **M7–M9:** Optimization of online update rules and preference learning.
- **M10–M12:** Final analysis, paper drafting, and submission.

References

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